8.3 Transversality

Transversal Intersections & Vector Fields

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Examples: Level Sets as Regular Submanifolds

Hypersurfaces determined by quadratic forms

Eg. Let A be a non-degenerate $(n+1)\times(n+1)$ real symmetric matrix. The quadratic form defines a smooth function φ_A:

$$\varphi_A(x_1,\ldots,x_{n+1}) = \sum_{i,j=1}^{n+1} a_{ij}\, x_i x_j.$$

When $s \neq 0$, $\varphi_A^{-1}(s)$ is either $\varnothing$ or an n-dimensional regular submanifold (a *hypersurface*).

The special linear group $SL(n, \mathbb{R})$

Eg. Take $n \times n$ real matrices as points in $\mathbb{R}^{n^2}$, i.e.

$$M_{n\times n}(\mathbb{R}) \cong \mathbb{R}^{n^2}.$$

Consider $f : M_{n\times n}(\mathbb{R}) \longrightarrow \mathbb{R}, \quad A \longmapsto \det(A)$. By the definition of det, f is a smooth map.

Claim. $SL(n, \mathbb{R})$ is an $(n^2 - 1)$-dim regular submanifold of $GL(n, \mathbb{R})$, an open subset of $\mathbb{R}^{n^2}$.

Idea

Show rank $f_{*,A} = 1$ at every $A \in GL(n, \mathbb{R})$ and apply the pre-image theorem.

$SL(n,\mathbb{R})$: proof

$GL(n,\mathbb{R}) = \left\{ M \,\middle|\, \det M \neq 0 \right\}$ is an open subset of $M_{n\times n}(\mathbb{R}) \cong \mathbb{R}^{n^2}$, hence a C^∞-manifold. Consider

$$f\big|_{GL(n,\mathbb{R})} : A \longmapsto \det(A) \in \mathbb{R}.$$

Given $A = (a_{ij}) \in GL(n,\mathbb{R})$, $\{a_{ij}\}$ is a local coordinate system about A, and

$$f_{*,A}\Big(\frac{\partial}{\partial a_{ij}}\Big|_A\Big) = \frac{\partial f}{\partial a_{ij}}(A)\,\frac{\partial}{\partial r} = A_{ij}\,\frac{\partial}{\partial r},$$

where A_{ij} is the minor of a_{ij}. If $\det A \neq 0$ these minors cannot all be zero, so $f_{*,A} : T_A GL(n,\mathbb{R}) \to T_{f(A)}\mathbb{R}$ has full rank, rank $= 1$. Hence $f^{-1}(\{1\}) = SL(n,\mathbb{R})$ is an (n^2-1)-dim regular submanifold of $GL(n,\mathbb{R})$.

A remark on the proof

Remark (Mei's book)

The author only checked $p \in SL(n, \mathbb{R})$: $f_{*,p}$ has full rank. This is enough, since if rank $f_* = \text{const}$ on $SL(n, \mathbb{R})$, it is constant on an open neighborhood of $SL(n, \mathbb{R})$.

Eg. A class of 2-dim surfaces. Let $F : \mathbb{R}^3 \to \mathbb{R}$, $F(x, y, z) = (x^2 + y^2)^2 + z^2$. When $(x, y, z) \neq 0$, $\text{rank}_{(x,y,z)} F = 1$. So for $r > 0$,

$$S_r = F^{-1}(\{r\})$$

is a regular submanifold of $\mathbb{R}^3 \setminus \{0\}$. Note that S_r is diffeomorphic to S^2.

Tangent Space of a Regular Submanifold

$T_pS = \ker f_{*,p}$

Eg. Let $f : M^m \to N^n$ be a smooth map with $\operatorname{rank} f \equiv l$ (identical everywhere). For $q \in N$, let $S = f^{-1}(\{q\}) \neq \varnothing$. Then $S \subset M$ is a regular submanifold, and $\forall\, p \in S$,

$$T_pS = \ker f_{*,p} = \left\{ w \in T_pM \;\middle|\; f_{*,p}(w) = 0 \right\}.$$

Proof.

For any tangent vector $w \in T_pS \subset T_pM$ and $h \in C^\infty(N)$,

$$f_{*,p}(w)(h) = w(h \circ f).$$

Since $f\big|_S \equiv \text{const} = q$, $\Rightarrow w\big(h \circ f\big|_S\big) = w\big(h(q)\big) \equiv 0$, so $w \in \ker f_{*,p}$ (equality by dimension). $\square$

Transversal Intersection

When a map meets a submanifold transversally

Definition (Transversal intersection)

Let $f: M \to N$ be a C^∞-map and $S \subset N$ a regular submanifold. If for any $p \in M$ with $f(p) = q \in S$ one has

$$f_{*,p}(T_pM) + T_qS = T_qN,$$

then f and S *intersect transversally*, denoted by $f \pitchfork S$. When $f(M) \cap S = \varnothing$, we also call f and S intersecting transversally.

If M, S are two regular submanifolds of N and for all $p \in M \cap S$,

$$T_pM + T_pS = T_pN,$$

we say M and S intersect transversally, denoted by $M \pitchfork S$.

The transversal intersection theorem

Theorem

Let $f\colon M \to N$ be a C^∞-map and $S \subset N$ a regular submanifold. If f intersects transversally with S, then $f^{-1}(S)$ is a regular submanifold of M, and

$$\dim M - \dim f^{-1}(S) = \dim N - \dim S.$$

Note

This generalizes the pre-image theorem: a *constant* rank condition on f is replaced by a transversality condition *along* S.

Regular values

Definition (Regular value)

Let $f : M \to N$ be a C^∞-map. If for every $p \in f^{-1}(\{q\})$ the differential $f_{*,p}$ is surjective, then $q \in N$ is called a *regular value* of f.

Lemma

Let $0 \in \mathbb{R}^k$ be a regular value of the C^∞-map $g : N \to \mathbb{R}^k$ and $S = g^{-1}(\{0\})$. Then $f : M \to N$ intersects transversally with $S \iff$ $0 \in \mathbb{R}^k$ is a regular value of $g \circ f$.

This reduces transversality of f with a level set to a regular-value check for $g \circ f$.

Proof of the lemma

If 0 is a regular value of $g \circ f$, then $\forall\, p \in (g \circ f)^{-1}(\{0\})$,

$$(g \circ f)_{*,p}(T_pM) = g_{*,f(p)} \circ f_{*,p}(T_pM) = T_0\mathbb{R}^k.$$

Since $T_{f(p)}S = \ker g_{*,f(p)}$ and $g_{*,f(p)}(T_{f(p)}N) = T_0\mathbb{R}^k$,

$$\begin{aligned} f_{*,p}(T_pM) + \ker g_{*,f(p)} &= T_{f(p)}N \\ \Rightarrow\ f_{*,p}(T_pM) + T_{f(p)}S &= T_{f(p)}N. \end{aligned}$$

Codimension

Definition (Codimension)

For a submanifold $S \subset N$, denote

$$\operatorname{codim} S = \dim N - \dim S.$$

Corollary (of the theorem)

In the transversal intersection theorem we can say

$$\operatorname{codim} f^{-1}(S) = \operatorname{codim} S.$$

Transversal pre-images keep the codimension of S.

Vector Fields

Smooth curves on a closed interval

Definition (Smooth curve)

A mapping $\sigma : [a, b] \to M$ is a *smooth curve* in M if σ extends to a C^∞-map of $(a - \varepsilon, b + \varepsilon) \to M$ for some $\varepsilon > 0$.

Note

If $\sigma : [a, b] \to M$ is a smooth curve, then its tangent vector

$$\dot{\sigma}(t) = \sigma_* \left(\frac{d}{dt}\Big|_t \right) \in T_{\sigma(t)} M$$

is well-defined for all $t \in [a, b]$.

Vector fields along a curve

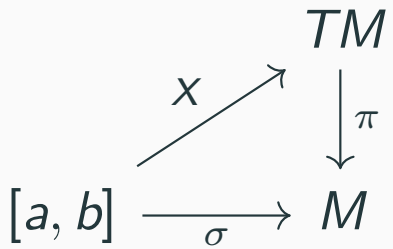
Definition (Vector field along a curve)

A smooth vector field X along a curve $\sigma : [a, b] \to M$ is a C^∞-map

$$X : [a, b] \longrightarrow TM$$

which *lifts* σ, i.e.

$$\pi \circ X = \sigma.$$



Vector fields on an open subset

Definition (Vector fields on an open subset of M)

A smooth vector field on an open set U in M is a C^∞-map $X : U \to TM$ s.t.

$$\pi \circ X = \mathrm{id}_U.$$

Sometimes we call a smooth vector field X on M a *section* of TM. The set of all smooth sections on U of TM is denoted by $\Gamma(U; TM)$. It is a vector space over $\mathbb{R}$ and also a $C^\infty(U)$-*module*.